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Loan Approval Prediction — Results and Findings Report

Colab Link: [🔗 Loan_Prediction_Complete_Colab.ipynb](#)

Executive summary

The final model classified all 400 held-out test applicants correctly (accuracy, precision, recall, F1 and ROC-AUC all 1.00), and 10-fold cross-validation on the full dataset gives 99.9% accuracy.

The dataset has 2,000 applicants and 7 columns. It contains no missing values, duplicates or invalid entries, and 17.1% of applications are approved. Approval follows an almost deterministic rule: Credit_Score above about 650, Income above about 50,000 and Loan_Amount below about 30,000. The data is very likely synthetic, so a score this high should not be expected on real lending data.

Data quality

The 2,000 rows × 7 columns contain no missing values, duplicates, invalid entries or outliers, so no rows were removed. Only 17.1% of applicants are approved (342 vs 1,658), so stratified splits and F1 were used.

What drives approval

- **Credit score:** none of the 1,255 applicants at 650 or below were approved.
- **Loan amount and income:** every approved loan is below about 30,000 and every approved applicant earns above about 50,000.
- **Little effect:** employment status (16.7–18.7% approval across groups), age and loan term.

Method

Stratified 80/20 split, four added affordability features (loan-to-income, monthly instalment, instalment-to-income, credit × income), and a leakage-free pipeline with imputation and scaling. Seven models were compared with 5-fold cross-validation, the best were tuned, and the cut-off (0.56) was set on out-of-fold data. The test set was used once.

Results

Model (5-fold CV)	Accuracy	F1
Random Forest, tuned (selected)	0.9988	0.9963
Gradient Boosting, tuned	0.9988	0.9963
Decision Tree	0.9988	0.9963
Logistic Regression	0.8938	0.7472

On the 400 test applicants the selected model made no errors: 332 true rejections, 68 true approvals, 0 false approvals, 0 false rejections. Precision, recall, F1, ROC-AUC and MCC are all 1.00. Ten-fold cross-validation on all rows gives accuracy 0.9990 ± 0.0030 .

Limitations

- The data looks synthetic and rule-generated, so real lending data will score lower.
- The test set has only 68 approved cases; the 10-fold figure is the more reliable estimate.
- Validate on real loan data before any practical use.